

Thesis Structure

Central Bank Dissonance and Asset Returns

Hierarchical Outline · Draft v0.1

Hierarchical outline with three levels of headings. Word counts are indicative for a master's-scale deliverable; the doctoral upgrade adds depth within the same architecture rather than new chapters.

Total body word count. $\approx$ 47,000 words for master's. $\approx$ 70,000–80,000 words for doctoral upgrade.

Chapter count. Nine substantive chapters plus front matter and back matter (including a timestamped pre-registration appendix).

Front Matter

- Title page
- Declaration of originality
- Abstract (300 words)
- Acknowledgements
- Table of contents
- List of figures
- List of tables
- List of abbreviations
- Notation glossary (R, D, τ , κ , etc.)

Chapter 1 — Introduction

≈ 4,000 words

1.1 Motivation

- 1.1.1 Central banks as multi-instrument institutions
- 1.1.2 The fragmentation of the existing literature
- 1.1.3 Three canonical cases of policy dissonance
 - 1.1.3.1 The 2013 Federal Reserve "taper tantrum"
 - 1.1.3.2 The European Central Bank 2011 cycle
 - 1.1.3.3 The 2023 US regional banking stress

1.2 Research Question

1.3 Approach in Brief

- 1.3.1 The structural decomposition
- 1.3.2 The dissonance metric
- 1.3.3 The fourteen-country empirical panel

1.4 Contributions

- 1.4.1 Theoretical
- 1.4.2 Empirical
- 1.4.3 Methodological

1.5 Scope and Limitations

- 1.5.1 The lens of monetary policy
- 1.5.2 What this thesis is not
- 1.5.3 The discretionary–systematic boundary

1.6 Structure of the Thesis

Chapter 2 — Literature Review

≈ 7,000 words

2.1 Frameworks for Central Bank Decomposition

- 2.1.1 The Svensson optimisation framework
- 2.1.2 Tinbergen and the multi-instrument tradition
- 2.1.3 The IMF Integrated Policy Framework
- 2.1.4 Borio and the BIS multi-mandate view

2.2 Reaction Functions

- 2.2.1 The Taylor rule and its descendants
- 2.2.2 Forward-looking and smoothed specifications
- 2.2.3 Open-economy and EM extensions
- 2.2.4 Estimated reaction functions for the panel countries

2.3 The Instrument Set

- 2.3.1 Forward guidance identification
- 2.3.2 Macroprudential policy measurement
- 2.3.3 Balance sheet policy and quantitative easing
- 2.3.4 FX intervention and capital flow management
- 2.3.5 Liquidity facilities

2.4 Transmission and Intermediary Frictions

- 2.4.1 The monetary transmission channels framework
- 2.4.2 Intermediary asset pricing
- 2.4.3 The dealer balance sheet and CIP basis
- 2.4.4 Sovereign-bank loops

2.5 Regime-Switching and Structural Break Detection

- 2.5.1 Markov-switching macroeconomic models
- 2.5.2 Structural break methods
- 2.5.3 Time-varying parameter VARs
- 2.5.4 Macro early-warning systems

2.6 Emerging Market Monetary Economics

- 2.6.1 Fear of floating and the trilemma
- 2.6.2 The global financial cycle
- 2.6.3 Sudden stops and capital flow reversals

2.7 Synthesis: Positioning the Contribution

- 2.7.1 What the literature decomposes
- 2.7.2 What the literature does not measure
- 2.7.3 The dissonance overlay

Chapter 3 — Theoretical Framework

≈ 6,000 words

3.1 The Structural Object

- 3.1.1 The directed graph $G = (N, E)$
- 3.1.2 The ten hierarchical levels
- 3.1.3 Tree, DAG, or general directed graph
- 3.1.4 Edge classes: transmission, feedback, peer
- 3.1.5 The pedagogical projection vs the formal object

3.2 Node Taxonomy

- 3.2.1 Agent nodes
- 3.2.2 Market nodes
- 3.2.3 Instrument nodes
- 3.2.4 Mathematical implications of species

3.3 Topology

- 3.3.1 Upstream and downstream sets
- 3.3.2 Transmission edges
- 3.3.3 Feedback edges
- 3.3.4 Peer edges and within-level relationships
- 3.3.5 The closed loop

3.4 The Consistency Model

- 3.4.1 The consistency function R_n
- 3.4.2 Dissonance as the violation of R_n
- 3.4.3 The normal-regime threshold τ_n
- 3.4.4 Theoretical grounding requirement

3.5 Dissonance Taxonomy

- 3.5.1 Intra-node dissonance
- 3.5.2 Inter-node dissonance
- 3.5.3 Vertical dissonance
- 3.5.4 Temporal dissonance

3.6 Country-Specific Parameterisation

- 3.6.1 Architecture universality
- 3.6.2 Sub-node activation
- 3.6.3 Numerical parameter specificity
- 3.6.4 Threshold non-linearities

3.7 Aggregation

- 3.7.1 The country-level composite $D_{c,t}$
- 3.7.2 Equal-weight scheme
- 3.7.3 Hierarchical scheme
- 3.7.4 PCA-extracted scheme
- 3.7.5 Theory-weighted scheme

3.8 Auxiliary Primitives

- 3.8.1 The modelling triad (state, dynamics, transmission)
- 3.8.2 The conditional transmission multiplier κ
- 3.8.3 Constraint binding
- 3.8.4 Frequency alignment
- 3.8.5 System regime characterisation

3.9 Summary of Definitions

Chapter 4 — Empirical Strategy

≈ 5,000 words

4.1 Sample Design

- 4.1.1 Country selection: the DM panel
- 4.1.2 Country selection: the EM panel
- 4.1.3 Time period and frequency
- 4.1.4 The DM/EM comparison as design feature

4.2 Node-Level Estimation

- 4.2.1 L1 reaction function estimation
- 4.2.2 L2 loss function and constraint indicators
- 4.2.3 L3 instrument dissonance metrics
- 4.2.4 L4 market dissonance metrics
- 4.2.5 L5 intermediary composites
- 4.2.6 L6 expectations gaps
- 4.2.7 L7 financial conditions composition

4.3 Composite Construction

- 4.3.1 Aggregation across nodes
- 4.3.2 Comparison of weighting schemes
- 4.3.3 Robustness across schemes as a result

4.4 Predictive Testing

- 4.4.1 Dependent variables: asset returns
- 4.4.2 Forecast horizons
- 4.4.3 Within-country regressions
- 4.4.4 Panel regressions
- 4.4.5 Out-of-sample design

4.5 Baseline Comparators

- 4.5.1 Statistical regime filters
- 4.5.2 Market-based stress indicators
- 4.5.3 Macro early-warning systems
- 4.5.4 Encompassing regression specification

4.6 Robustness Strategy

- 4.6.1 Drop-one-node analysis
- 4.6.2 Drop-one-country analysis
- 4.6.3 Alternative threshold calibrations
- 4.6.4 Crisis vs non-crisis sub-samples

4.7 Pre-Registration

Chapter 5 — Data

≈ 4,000 words

5.1 Database Architecture

- 5.1.1 Schema and storage
- 5.1.2 Country-time-frequency harmonisation
- 5.1.3 Version control

5.2 Data Sources by Node Level

- 5.2.1 Macroeconomic data (L1, L2, L10)
- 5.2.2 Central bank action data (L3)
- 5.2.3 Market data (L4, L9)
- 5.2.4 Intermediary data (L5)
- 5.2.5 Expectations data (L6)
- 5.2.6 Financial conditions (L7)

5.3 Data Quality Assessment

- 5.3.1 Coverage matrix by country and node
- 5.3.2 Frequency mismatches
- 5.3.3 Proxy decisions and justifications

5.4 Limitations and Scope Restrictions

- 5.4.1 NBFI data in emerging markets
- 5.4.2 Dealer-balance-sheet measurement
- 5.4.3 Communication data in non-English-language regimes

Chapter 6 — South Africa Pilot

≈ 5,000 words

6.1 Rationale for South Africa as Pilot

6.2 Node-by-Node Implementation

- 6.2.1 Reaction function: Aron-Muellbauer specification
- 6.2.2 Instrument vector: SARB historical record
- 6.2.3 Communication: NLP on MPC statements
- 6.2.4 Intermediaries: the Big Four banks and the sovereign
- 6.2.5 Markets: the R-bond curve and ZAR

6.3 The South African Dissonance Time Series, 2005–2024

- 6.3.1 Descriptive overview
- 6.3.2 Identified regime events
- 6.3.3 Composite signal under each weighting scheme

6.4 Sanity Checks

- 6.4.1 The 2008–09 global financial crisis
- 6.4.2 The 2013 taper tantrum
- 6.4.3 The 2015–16 commodity cycle
- 6.4.4 The 2020 COVID shock
- 6.4.5 The 2022–23 hiking cycle

6.5 Preliminary Predictive Analysis

6.6 Framework Adjustments Pre-Panel

- 6.6.1 Implementation choices not in the pre-registration
- 6.6.2 Justifications and audit trail

Chapter 7 — Panel Results

≈ 8,000 words

7.1 Cross-Country Dissonance Patterns

- 7.1.1 Descriptive statistics by country
- 7.1.2 Node-level firing frequencies
- 7.1.3 Common factor analysis across the panel

7.2 The Developed Market Sub-Panel

7.3 The Emerging Market Sub-Panel

7.4 Direct DM–EM Comparison

- 7.4.1 Frequency and magnitude differences
- 7.4.2 Node-level firing differences
- 7.4.3 Asset-class response differences

7.5 Predictive Regressions

- 7.5.1 Local-currency sovereign bond returns
- 7.5.2 Exchange rate returns
- 7.5.3 Equity index returns
- 7.5.4 Sovereign CDS returns

7.6 Out-of-Sample Performance

- 7.6.1 Expanding-window scheme
- 7.6.2 Comparison to in-sample fit

7.7 Encompassing Regressions Against Baselines

- 7.7.1 Against Markov-switching regime probability
- 7.7.2 Against FCI z-scores
- 7.7.3 Against the excess bond premium analogue
- 7.7.4 Against the Kaminsky-Reinhart signals approach

7.8 Robustness

- 7.8.1 Across weighting schemes
- 7.8.2 Drop-one-node
- 7.8.3 Drop-one-country
- 7.8.4 Alternative thresholds

Chapter 8 — Mechanism and Case Studies

≈ 5,000 words

8.1 Conditional Sharpe Analysis

- 8.1.1 Equilibrium-regime performance
- 8.1.2 Transition-regime performance
- 8.1.3 The regime-change-detector hypothesis

8.2 Node-Level Attribution

- 8.2.1 Which nodes carry the most predictive weight
- 8.2.2 Decomposition of $D_{c,t}$ contribution to forecast accuracy

8.3 Asset-Level Attribution

- 8.3.1 Mapping fired nodes to asset responses
- 8.3.2 Differentiating across asset classes

8.4 Case Studies

- 8.4.1 The 2013 US taper tantrum
- 8.4.2 The ECB 2011 cycle
- 8.4.3 The 2023 US regional banking stress
- 8.4.4 Turkey 2018
- 8.4.5 South Africa 2020 and 2022–23
- 8.4.6 A null case: an event where dissonance did not predict

8.5 Limits of the Framework

- 8.5.1 What the framework misses
- 8.5.2 False positives and false negatives

Chapter 9 — Discussion and Conclusion

≈ 3,000 words

9.1 Summary of Findings

9.2 Contributions Revisited

- 9.2.1 Theoretical contribution

- 9.2.2 Empirical contribution
- 9.2.3 Methodological contribution

9.3 Implications for Monetary Economics

9.4 Implications for Empirical Asset Pricing

9.5 Implications for Investment Practice

9.6 Limitations

9.7 Future Research

- 9.7.1 Extending to developed market intermediary detail
- 9.7.2 Structural interpretation as monetary shock identification
- 9.7.3 Cross-level dissonance as a standalone research programme
- 9.7.4 Operationalisation for live portfolio management

Back Matter

References

Appendix A — Mathematical Definitions and Proofs

Appendix B — Data Sources, Transformations, and Coverage Matrix

Appendix C — Pre-Registration Document (timestamped)

Appendix D — Country-Specific Reaction Function Estimates

Appendix E — Node-Level Dissonance Time Series, Full Panel

Appendix F — Additional Robustness Tables

Appendix G — Code Repository and Replication Instructions

Structural choices worth flagging

1. Chapter 6 (SA Pilot) stands alone as a chapter. This treats the pilot as a substantive deliverable in its own right rather than burying it in the appendix. The advantage is a complete case study that can be excerpted as a standalone paper. The cost is some redundancy with Chapter 7.

2. Chapter 8 includes a deliberate null case study. Asymmetric reporting — only showing cases where the framework "works" — would weaken the thesis. Including a case where dissonance did not predict (or fired falsely) is what makes the empirical claim credible.

3. Appendix C is the pre-registration document, timestamped. Unusual for an economics thesis but it's the cleanest way to defend against post-hoc-tinkering critiques and pre-empt the most likely examiner concern.

Next: Section 3.1 — the Structural Object — where the directed-graph-vs-tree question is formally resolved.